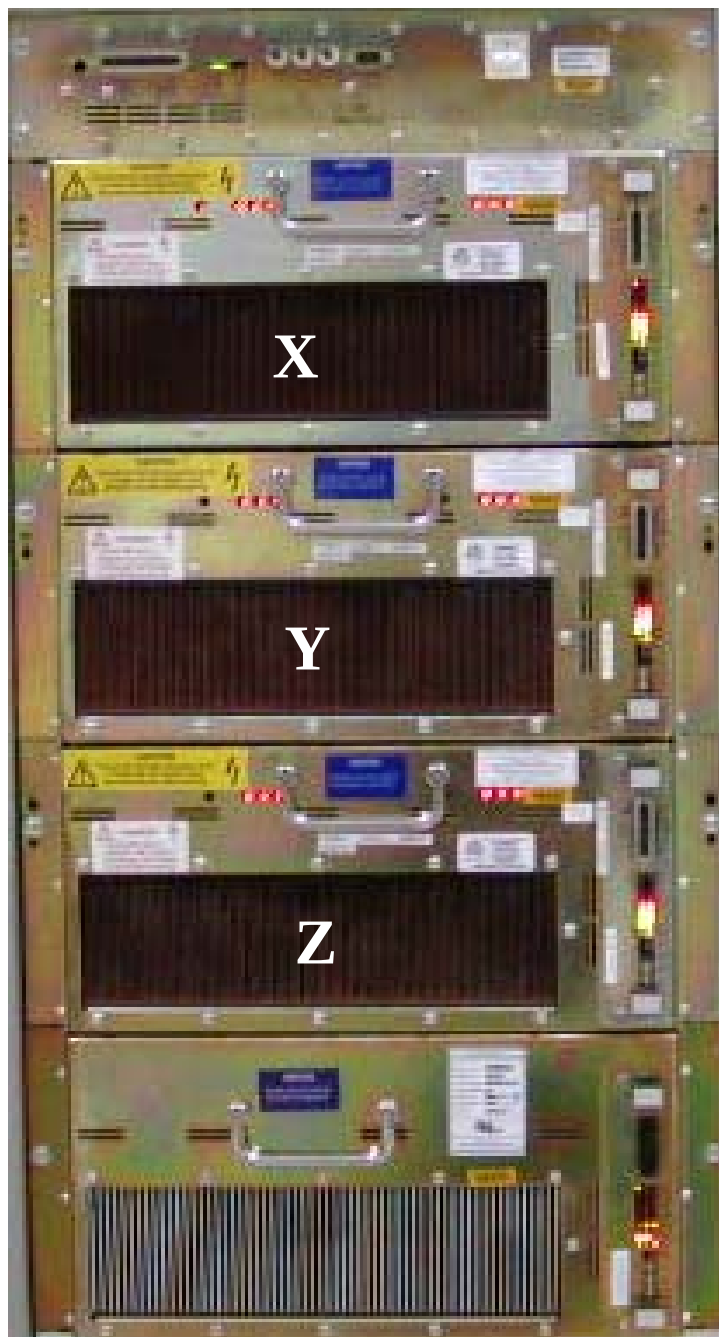




The **Gradient Processor 3 (GP3)** interfaces the MGD STIF Board with the HFAs, the SGAPS, and the Gradient Switch. High speed fiber optic data for the X, Y, and Z gradient amplifiers is received and translated into control signals for the HFAs. Fault handling, status and mode control, eddy-current compensation, and diagnostics are also handled by the GP.

The **High Fidelity Gradient Amplifiers (HFAs)** generate current for the Gradient Coils based on control signals received from the GP3. Current is generated using a pulse-width modulated (PWM) output acting on four DC voltages, provided by the SGAPS. The Gradient Coil current is sensed and returned to the GP3 for optimal error control.

The **SGA Power Supply (SGAPS)** provides four DC voltages to each HFA which are switched on and off to generate current on the Gradient Coil.